

Deep Learning — Assignment

Building, Breaking and Fixing a Neural Network

Summary Report

Fashion-MNIST · Feed-forward Neural Networks

Student name	[your name]
Roll number	23F-0611
Course	[course name / code]
Instructor	[instructor name]
Notebook	23f-0611-dlp-4-5.ipynb
Platform	Kaggle Notebook, GPU T4 x2
Dataset	Fashion-MNIST (zalando-research/fashionmnist)
Random seed	42 (set for Python, NumPy, PyTorch and CUDA)
Date	[submission date]

Every number in this report is printed by a cell in the submitted notebook.

Deep Learning for Perception

Assignment 01

23F-0611

1. Final test score

The tuned model from Part 7 was evaluated once on the untouched 10,000-image test set.

Metric	Test-set value
Accuracy	0.8771 (87.71 %)
Macro precision	0.8783
Macro recall	0.8771
Macro F1	0.8776

Part 2 baseline test accuracy: 0.8800. Change after tuning: -0.29 percentage points. This is reported as measured; the tuning did not improve on the baseline.

2. Configuration that produced it

Item	Value
Architecture	784 $\rightarrow$ 256 $\rightarrow$ 256 $\rightarrow$ 10, ReLU hidden layers, softmax output
Optimiser	Adam, learning rate 0.003
Dropout rate	0.0 (selected by the search)
L2 weight decay	1e-4 (best value from Part 6)
Early stopping	patience 5 on validation loss — stopped at epoch 14
Batch size	128
Training data	Full 48,000-sample training split
Selection method	Random search over 12 configurations, 5-fold cross-validation
Cross-validation score	0.8522 $\pm$ 0.0017

3. The single change that helped most

Increasing the amount of training data. Holding the Part 5 network and every other setting fixed, raising the training set from 2,000 to 20,000 samples cut the generalisation gap from 0.144 to 0.081 while training accuracy stayed at 0.968 — the only method tested that reduced the gap at no cost in fit — and it raised validation accuracy by 6.9 percentage points, the largest gain of any method in Part 6.

Method (Part 6)	Train acc	Val acc	Gap
Baseline, 2,000 samples	0.9630	0.8187	0.1442
More data, 20,000 samples	0.9684	0.8879	0.0805
Dropout 0.6 (runner-up)	0.8865	0.8311	0.0554

Dropout at rate 0.6 gave a slightly larger reduction in the gap, but it cost 7.7 points of training accuracy and gained only 1.2 points of validation accuracy, so more data remains the better change.

4. Note on the negative result

The cross-validation budget was short (10 epochs per fold on a 10,000-sample subset), which favours configurations that learn quickly rather than those that finish best; the winning configuration used a high learning rate and no dropout. Combined with a baseline already near the ceiling for fully connected networks on Fashion-MNIST, this left almost no headroom, and the 0.29-point shortfall corresponds to 29 images out of 10,000.